

Multigranulation Relative Entropy-Based Mixed Attribute Outlier Detection in Neighborhood Systems

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Abstract—Outlier detection is widely used in many fields, such as intrusion detection, credit card fraud detection, medical diagnosis, and so on. Existing outlier detection algorithms are mostly designed for dealing with numeric or categorical attributes. However, data usually exist in the form of mixed attributes in real-world applications. In this article, we propose a novel mixed attribute outlier detection method based on multigranulation relative entropy by employing the neighborhood rough set. First, the neighborhood system is constructed by optimizing the mixed distance metric and the radius of the statistical value. Second, the neighborhood entropy is introduced as an uncertainty measure of data. Furthermore, the three kinds of multigranulation relative entropy-based matrices are defined by three kinds of attribute sequences, and the multigranulation relative entropy-based outlier factor is integrated to indicate the outlier degree of every object. Based on the proposed outlier detection model, the corresponding algorithm is designed. Finally, the proposed algorithm is compared with other nine algorithms through experiments on public data. The experimental results show that the proposed technique is adaptive and effective.

Index Terms—Mixed attribute, multigranulation, neighborhood rough set theory, outlier detection, relative entropy.

I. INTRODUCTION

OUTLIER detection is one of the most important research directions in data mining, whose purpose is to find out the objects with the behavior very different from other objects in the data [1]. In some studies, outliers are eliminated as abnormalities [2]. However, there are many applications for

outlier detection, such as intrusion detection, credit card fraud, interesting sensor events, medical diagnosis, law enforcement, Earth science, and so on [3]–[5].

In recent decades, outlier detection has received extensive attention from researchers, and many outlier detection methods have been proposed. Existing outlier detection methods mainly include probability and statistical models [6], proximity-based models [7]–[9], and high-dimensional outlier detection [10]. Among them, the idea of proximity-based models is to treat objects that deviate from the rest of the data based on a distance metric as outliers. It mainly includes three methods: 1) cluster-based methods [7]; 2) distance-based methods [8]; and 3) density-based methods [9].

For cluster-based methods, the first step is to apply a clustering algorithm to calculate the dense regions of a data. The outlier factors of the data objects are then calculated by measuring the adaptability to different clusters. One of its important advantages is that it is relatively fast compared with distance-based methods. The main weakness of this method is that they may not always provide the required details in a smaller data set. Another problem with cluster-based methods is that outlier factors are highly variable between different random executions or parameter selections.

The distance-based method is a widely used outlier detection algorithm in various data fields, which define an outlier factor based on the nearest neighbor distance. Compared with cluster-based methods, it has many merits, because the analysis is finer in granularity. For instance, it is better than the cluster-based algorithm in identifying noise and anomalies. Besides, it can also find isolated outliers such as cluster-based methods. However, most distance-based algorithms are designed by employing Euclidean metric. In practical applications, Euclidean distance may not be optimal for detecting outliers of categorical (nominal) or mixed (hybrid or heterogeneous) attribute data.

Density-based methods apply the number of objects in a particular region of the space to define outliers. According to the local neighborhood density, the local outlier factor (LOF) of each data object is constructed. Then, the data object with high LOF value is identified as an outlier. A drawback is that it is very sensitive to parameter selection and is also constructed by using Euclidean metric.

The rough set theory (RST) provides mathematical tools for insufficient and incomplete data-based classification learning. In particular, it provides a formal framework for categorical

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attribute reduction, feature selection, classification learning, information fusion, intrusion detection, etc., [11]–[18]. As mentioned above, most of distance-based and density-based methods apply Euclidean metric to compute the distance between data objects. There is no inherent distance metric between categorical attribute values, so it cannot effectively process data containing categorical attributes. In addition, after analysis, we find that most outlier detection methods basically use a deterministic way to represent and process data, and do not consider the uncertainty and incompleteness of the data set. However, a large amount of uncertainty and incomplete data exist in data sets collected from real life. For these reasons, many detection methods using RST have also been put forward [19]–[27]. For example, Chen *et al.* [19] put forward a granular computing (GrC)-based detection method. Jiang *et al.* [20] extended two detection approaches, including sequence based and distance based by employing RST. Jiang *et al.* [21] again extended a novel detection approach through using information entropy in rough set. Albanese *et al.* [23] put forward an outlier detection method of spatiotemporal data by utilizing a novel RS method. In [24], a computationally efficient algorithm was introduced based on rough set for detecting outlier in large volumes of information. A novel detection method was introduced from the viewpoints of GrC and RST in [25]. Recently, Jiang *et al.* [26] proposed an outlier detection algorithm based on approximation accuracy entropy within the framework of rough set. Singh and Pamula [27] proposed an outlier detection method for large-scale data streams using rough set.

The above-mentioned methods have proved the validity of outlier detection based on RST. However, these methods build mathematical models based on equivalence relations, which are only suitable for categorical attribute rather than numeric (numerical) attribute data. In these detection models, discretization is needed when processing numerical attribute data, which increases the time consumption and significant information loss. In order to deal with mixed data type, neighborhood relation is considered in RST. Lin [28] introduced neighborhood and neighborhood relation to RST. Yao [29] studied the properties of neighborhood approximation spaces. Hu *et al.* [30]–[32] proposed a neighborhood rough set model and gave some feature selection algorithms. The neighborhood rough set is a powerful data mining tool for numerical or mixed attribute feature selection, classification, reasoning of uncertainty, etc., [32], [33]. Nevertheless, mixed attribute outlier detection is rarely reported in the neighborhood rough set model in recent years [34]–[37]. For instance, Chen *et al.* [34] introduced the neighborhood model as a uniform framework to construct neighborhood outlier detection. However, this detection method involves many parameters of the neighborhood radius and needs to be artificially set, so it has strong subjectivity and randomness. Recently, based on neighborhood entropy, Yuan *et al.* [35] proposed a hybrid data-driven outlier detection method. However, in this model, the information fusion mainly uses the integration technology of single attribute, which may have a weak performance on some data sets. Wang and Li [36] proposed an outlier detection method based on a weighted network model for mixed-value

data sets in neighborhood rough sets. The method based on neighborhood rough set is used to detect prototype outliers in [37].

GrC is a general computing theory, which is used to build efficient computing models for complex applications using granules. As a new information processing concept and computing paradigm, GrC is a conceptual framework for studying many problems in data mining and pattern recognition [38]–[41]. Herein, Qian *et al.* [39] extended Pawlak's rough set model to a multigranulation rough set model by using multiequivalence relations on the universe. Furthermore, the information entropy proposed by Shannon [42] is a competitive mechanism for measuring uncertainty. In order to measure the uncertainty of rough set, many researchers apply information entropy to rough set and propose many different information entropy models [43]–[46]. However, these information entropy are mainly based on the classical rough set. For this reason, Hu and Yu [47] proposed a neighborhood entropy.

In this article, we construct a novel multigranulation relative entropy-based outlier detection (MREOD) method. First, the method optimizes the heterogeneous neighborhood distance metric and the neighborhood radius of the statistical value to construct a new neighborhood information system (NIS). Then, the relative entropy is defined by employing neighborhood entropy proposed in [47] to describe the uncertain information of data objects. Second, the sequence of attribute is defined according to the idea of sequences and multigranulation [20], [39]. Subsequently, in order to form multineighborhood relations, two kinds of sequences of attribute subset (i.e., the forward and reverse sequences) are defined based on the sequence of attribute. Based on the three kinds of sequences, we construct three kinds of sets of multigranulation neighborhood entropy sequences. Correspondingly, the three kinds of multigranulation relative entropy-based matrices are obtained. Finally, the multigranulation relative entropy-based outlier factor (MREOF) is calculated to characterize the outlier degree of each object. Moreover, the relevant MREOD algorithm is designed.

As a result, the main contributions of this article are as follows.

- 1) A new NIS is constructed by defining a novel heterogeneous neighborhood distance metric.
- 2) A new neighborhood entropy is employed to define three sequences.
- 3) The outlier detection model is constructed by fusing the idea of sequence and multigranulation.
- 4) Algorithm MREOD can be applied to three kinds of data, including categorical, numerical, and mixed attribute data.
- 5) The experimental results show that the proposed method has better adaptability and effectiveness than the existing nine outlier detection methods.

The remainder of this article is structured as follows. In Section II, we review some preliminaries of neighborhood rough set and neighborhood entropy that are relevant to this article. In Section III, we propose a mixed attribute outlier detection method based on multigranulation relative entropy.

Experimental results and analyses are given in Section IV. Finally, Section V concludes this article.

II. PRELIMINARIES

In this section, some knowledge of NIS and neighborhood entropy are reviewed in neighborhood rough set [31], [47].

An information system (IS) without decision can be expressed formally as a quadruple $IS = (U, C, V, f)$, where $U = \{x_1, x_2, \dots, x_n\}$ is a nonempty finite set of objects called universe; C is a nonempty finite set of conditional attributes; $V = \bigcup_{c \in C} V_c$ is the union of attribute domain V_c ; and $f : U \times C \rightarrow V$ is an information function: $\forall x \in U$ and $c \in C$, s.t., $f(x, c) \in V_c$.

A. Distance Metric

For $\forall x, y, z \in U$, a distance function on $B \subseteq C$ is denoted as $d_B : U \times U \rightarrow [0, +\infty)$, which satisfies:

- 1) $d_B(x, y) \geq 0$, $d_B(x, y) = 0$ if and only if $x = y$;
- 2) $d_B(x, y) = d_B(y, x)$;
- 3) $d_B(x, z) \leq d_B(x, y) + d_B(y, z)$.

There are many distance metric, such as the Minkowski distance (including Manhattan distance, Euclidean distance, and Chebychev distance), heterogeneous Euclidean-overlap metric (HEOM), value difference metric (VDM), heterogeneous VDM (HVDM), and interpolated VDM (IVDM) [48]. For example, HEOM on C is defined as

$$HEOM(x, y) = \sqrt{\sum_{j=1}^m \omega_{c_k} d_{c_k}^2(x, y)} \quad (1)$$

where

$$d_{c_k}(x, y) = \begin{cases} 1, & \text{if the } f(x, c_k) \text{ or } f(y, c_k) \text{ is unknown} \\ 0, & \text{if } c_k \text{ is a categorical attribute} \\ & \text{and } f(x, c_k) = f(y, c_k) \\ 1, & \text{if } c_k \text{ is a categorical attribute} \\ & \text{and } f(x, c_k) \neq f(y, c_k) \\ \frac{|f(x, c_k) - f(y, c_k)|}{\max_{c_k} - \min_{c_k}}, & \text{if } c_k \text{ is a numerical attribute.} \end{cases}$$

ω_{c_k} is the weight of attribute c_k , and $\max_{c_k}$ and $\min_{c_k}$ represent the maximum and minimum values of attribute c_k , respectively.

From the above definition, we can see that HEOM can handle both numerical and categorical attributes, and can also be used when some attribute values are unknown.

B. Neighborhood Information System

Because equivalence relation and equivalence class are only suitable for solving categorical attribute data sets, on the basis of distance metric, the neighborhood radius ε can be introduced into granulate U to form neighborhood and neighborhood class, so as to obtain a NIS.

Definition 1: For $\forall B \subseteq C$, the neighborhood $n_B(x)$ of x with respect to B is

$$n_B(x) = \{y | d_B(x, y) \leq \varepsilon_B, y \in U\}. \quad (2)$$

The $n_B(x)$ includes objects, whose distance between x on the B is not more than ε , so it is an neighborhood granule based on the distance.

Furthermore, $nr_B = \{(x_i, x_j) | x_i, x_j \in U, x_j \in n_B(x_i)\}$ is defined as B -neighborhood relation on U . Obviously, neighborhood relation satisfies reflexivity and symmetry, so it is a kind of similarity relations. Especially, nr_B is the equivalence relation on U if $\varepsilon_B = 0$, this case is suitable for categorical attribute data; nr_B is the similarity relation on U if $\varepsilon_B > 0$, this case is applicable to numerical attribute data.

$Cover(nr_B)$ denotes a covering of U , which is called a neighborhood knowledge on U . Let $NR_C = \{nr_B | B \subseteq C\}$ denote all the neighborhood relations on U , and the quadruple $NIS = (U, NR_C, V, f)$ is called as a NIS.

Obviously, different distance metrics and different ε can obtain different NIS. When $\varepsilon_B = 0$, the neighborhood class of x is degenerated into the equivalence class of x , i.e., $n_B(x) = [x]_B$. The NIS degenerates to the classical IS.

C. Neighborhood Entropy

The neighborhood entropy proposed by Hu *et al.* can deal with both categorical and numerical attribute data [47], which is given as follows.

Definition 2 [47]: The neighborhood uncertainty of x_i on B is computed by

$$NH^{x_i}(B) = -\log_2 \frac{|n_B(x_i)|}{|U|} \quad (3)$$

where $|\bullet|$ denotes the cardinality of a set $\bullet$.

Definition 3 [47]: The neighborhood entropy $NH(B)$ on B is

$$NH(B) = -\frac{1}{|U|} \sum_{i=1}^{|U|} \log_2 \frac{|n_B(x_i)|}{|U|}. \quad (4)$$

In the above definition, neighborhood entropy is calculated from the average uncertainty of neighborhood on each object. It can measure the uncertainty of the neighborhood rough set. If nr_B degenerates into a crisp equivalence relation, the proposed neighborhood entropy is the same as the classical Shannon entropy. This lays a theoretical foundation for the construction of the mixed attribute outlier detection method.

III. PROPOSED OUTLIER DETECTION METHOD

A mixed attribute outlier detection method based on multi-granulation relative entropy is presented in this section, which mainly involves problem analysis, detection system, detection method, and detection algorithm.

A. Problem Analysis

As mentioned in the introduction, the shortcomings of some existing methods are summarized as follows.

- 1) Distance-based methods and density-based methods are usually applied to numeric attributes.
- 2) Rough set-based methods are only suitable for categorical attributes.
- 3) The method proposed in [35] mainly uses a single attribute integration technique, so the performance on some data sets may be weak.

In order to illustrate the limitation of algorithm NIEOD in [35], an example is given as follows.

TABLE I
IS AND THE CORRESPONDING DESCENDING NEOF VALUES

U	c_1	c_2	U	$NEOF$
x_1	0.2	0.8	x_7	0.7714
x_2	0.3	0.7	x_1	0.6942
x_3	0.4	0.6	x_5	0.6769
x_4	0.5	0.5	x_8	0.6623
x_5	0.6	0.4	x_2	0.6623
x_6	0.7	0.3	x_6	0.6568
x_7	0.9	0.1	x_4	0.6532
x_8	0.8	0.8	x_3	0.6006

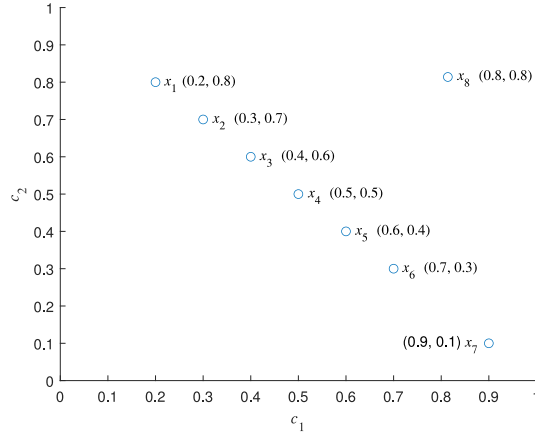


Fig. 1. Distribution of data in Table I.

Example 1: An information system $IS = (U, C, V, f)$ is shown in Table I, where $U = \{x_1, x_2, \dots, x_8\}$ and $C = \{c_1, c_2\}$. Let $\lambda = 1$, the related descending NEOF values are listed in the last column of Table I. Given threshold $\mu = 0.69$, there are only $NEOF(x_7) > \mu$ and $NEOF(x_1) > \mu$. Therefore, the outlier set is $OS = \{x_1, x_7\}$. Since $NEOF(x_8) < \mu$, x_8 is not considered to be an outlier. However, the data distribution depicted in Fig. 1 shows that x_1 and x_8 are somewhat objective outliers. We can see that algorithm NIEOD does not correctly recognize x_8 as an outlier, resulting in a weakness.

Considering the above problem analyses, this section constructs a mixed attribute outlier detection method based on multigranulation relative entropy.

B. Detection System

In this section, a NIS is constructed for outlier detection, which mainly involves normalization preprocessing, selection of neighborhood metric, and standard setting of neighborhood radius.

The dimension of data in numerical features is usually different. In order to avoid the effect of data dimension on the data mining results, the original numerical attribute data are normalized before data processing. The min-max normalization method is used in this article, which is as follows:

$$F(f(x_i, c_k)) = \frac{f(x_i, c_k) - \min_{c_k}}{\max_{c_k} - \min_{c_k}}. \quad (5)$$

After normalization, the range of attribute values for these numerical attributes is $[0, 1]$.

In order to effectively deal with heterogeneous attribute data, a HEOM method is proposed in [48]. Similarly, the heterogeneous Manhattan-overlap metric (HMOM) can be defined as follows.

Definition 4 (HMOM): For $\forall x, y \in U$, let $B = \{c_{k_1}, c_{k_2}, \dots, c_{k_l}\} (1 \leq l \leq m) \subseteq C$. The HMOM $HMOM_B(x, y)$ of x and y on B is defined as

$$HMOM_B(x, y) = \sum_{h=1}^l d_{c_{k_h}}(x, y) \quad (6)$$

where

$$d_{c_{k_h}}(x, y) = \begin{cases} 0, & \text{if } c_{k_h} \text{ is categorical and } f(x, c_{k_h}) = f(y, c_{k_h}) \\ 1, & \text{if } c_{k_h} \text{ is categorical and } f(x, c_{k_h}) \neq f(y, c_{k_h}) \\ |f(x, c_{k_h}) - f(y, c_{k_h})|, & \text{if } c_{k_h} \text{ is numerical.} \end{cases}$$

Most of the data in real life are heterogeneous. From the above definition, we can see that HMOM can not only deal with the numeric data but also can cope with the complex data with the combination of numeric and categorical attributes at the same time. Therefore, this article adopts HMOM as a neighborhood distance metric. It lays the foundation for expanding the scope of application.

The acquisition of neighborhood involves neighborhood radius, and the neighborhood radius is usually determined by experts according to experience [31], [34], [47]. For example, if the setting method of neighborhood radius in [34] is adopted, the parameters involved will be as many as the number of attributes m . Obviously, it is subjective and random, which leads to the sensitivity of the algorithm to parameter selection. Besides, reducing the sensitivity of the algorithm to the parameters specified by experts is the objective basis for improving the accuracy of the algorithm. A more reasonable method is to determine the neighborhood radius by fusing the actual value distribution information of attribute value with the parameters given by an expert. Therefore, the neighborhood radius of the object x with respect to attribute c_k is set as [35]

$$\varepsilon_{c_k} = \begin{cases} \frac{\text{std}(c_k)}{\delta}, & \text{if } c_k \text{ is numerical} \\ 0, & \text{if } c_k \text{ is categorical} \end{cases} \quad (7)$$

where $\text{std}(c_k)$ denotes the standard deviation of the numerical attribute values, while δ is a parameter for radius adjustment.

The standard deviation $\text{std}(c_k)$ is used to measure the degree of dispersion of a data set. The neighborhood radius is more than $\text{std}(c_k)$, if $\delta < 1$; the neighborhood radius is equal to $\text{std}(c_k)$, if $\delta = 1$; and the neighborhood radius is less than $\text{std}(c_k)$, if $\delta > 1$.

In summary, $\text{std}(c_k)$ is an important factor for adjusting the neighborhood radius, so more scientific statistics and objectivity are added. It lays the foundation for the adaptability and effectiveness of the follow-up detection methods.

In addition, the neighborhood radius on B of an object is set to $\varepsilon_B = \sum_{h=1}^l \varepsilon_{c_{k_h}}$. When numeric and categorical attributes

coexist, suppose $B_1, B_2 \subseteq C$ be numeric and categorical attributes, respectively. Then, the neighborhood of x is defined as follows [31].

- 1) $n_{B_1}(x) = \{y | \text{HMOM}_{B_1}(x, y) \leq \varepsilon_{B_1}, y \in U\}$.
- 2) $n_{B_2}(x) = \{y | \text{HMOM}_{B_2}(x, y) = 0, y \in U\}$.
- 3) $n_{B_1 \cup B_2}(x) = \{y | \text{HMOM}_{B_1}(x, y) \leq \varepsilon_{B_1} \text{ and } \text{HMOM}_{B_2}(x, y) = 0, y \in U\}$.

Through the above normalization, HMOM metric, ε , $n_B(x)$, nr_B , and $\text{Cover}(nr_B)$, then NIS is established. In the next section, the MREOF is constructed to indicate the outlier degree of each object x in U .

C. Detection Method

Hu and Yu proposed a neighborhood entropy [47], which is calculated from the average uncertainty of each object's neighborhood in U . But each object may have the same neighborhood cardinality, which may lead to redundancy of uncertainty information. Thus, in order to construct the outlier detection model effectively, the neighborhood entropy is calculated by the degree to which the sum of the uncertainties of different neighborhoods belongs to U , which is defined as follows. Given $\delta > 0$, for $\forall nr_B \in NR_C$, let $DN_U(nr_B) = \{N_1, N_2, \dots, N_s\}$ denote s different neighborhoods on U .

Definition 5: The neighborhood entropy $NE(B)$ on nr_B is defined as

$$NE(B) = -\frac{1}{|U|} \sum_{r=1}^s \log_2 \frac{|N_r|}{|U|}. \quad (8)$$

In the above definition, if $DN_U(nr_B) = U$, then the neighborhood entropy $NE(B)$ achieves the minimum value 0. If $DN_U(nr_B) = \{\{x_1\}, \dots, \{x_n\}\}$, then $NE(B)$ achieves the maximum value $\log_2 n$. So we have $0 \leq NE(B) \leq \log_2 n$.

In order to construct the multigranulation relative entropy-based outlier factor, we put forward a new concept of relative entropy based on neighborhood entropy, which indicates uncertainty of every object in U .

Definition 6: For $\forall x \in U$, let $DN_{U-\{x\}}(nr_B) = \{N'_1, N'_2, \dots, N'_s\}$ denote s' different neighborhoods on $U - \{x\}$. Then, the relative entropy $RE_B(x)$ of x with respect to nr_B is defined as

$$RE_B(x) = \begin{cases} \frac{NE_x(B)}{NE(B)}, & 0 \leq \frac{NE_x(B)}{NE(B)} \leq 1 \\ 1, & \text{otherwise} \end{cases} \quad (9)$$

where $NE_x(B) = -(1/|U - \{x\}|) \sum_{r=1}^{s'} \log_2(|N'_r|/|U - \{x\}|)$ denotes the neighborhood entropy on nr_B when removing x from U .

$NE_x(B)$ can measure the uncertainty of x . When we delete x from U , on the one hand, if $NE_x(B)$ decreases greatly, then we may consider the uncertainty of x is high. On the other hand, if $NE_x(B)$ changes little or even increases, then we may consider the uncertainty of x is low. Then, based on $NE_x(B)$ and its semantics, $RE_B(x)$ gives a measure for the uncertainty of x . Concretely, the lower the relative entropy $RE_B(x)$ of x , the higher the uncertainty of x .

In addition, the weight function positively determines the outlier factor and detection result but it is an empirical function. In order to get more accurate outlier detection results,

based on a large number of experiments, the specific weight is defined as follows.

Definition 7: The weight function $W_B(x)$ of x on nr_B is defined by

$$W_B(x) = \sqrt[3]{\frac{|n_B(x)|}{|U|}}. \quad (10)$$

The basic idea behind the above weight function setting is that the detection method always focuses on the minority group, because objects belonging to minority group are more possible to be outliers. Therefore, objects that belong to the minority group should have lower weights. In (10), if there are few objects in the neighborhood of x , then x has a small percentage in U , corresponding to a minority group and a lower weight. Obviously, we have $W_B(x) \in (0, 1]$.

From the above Definitions 6 and 7, we can see that different relative entropy and weight functions can be obtained via different neighborhood relations. Each $B \subseteq C$ determines a neighborhood relation nr_B , which is also deemed as the neighborhood knowledge in NIS. Hence, there are $2^{|C|}$ different neighborhood relations, which can be used to build the knowledge base. Clearly, we can obtain more data information by using $2^{|C|}$ neighborhood relations to calculate $RE_B(x)$ and $W_B(x)$ of each object. However, it is unfeasible to employ all these relations to compute $RE_B(x)$ and $W_B(x)$. Its time complexity will be exponential on $|C|$. Therefore, we employ an effective strategy. That is, we construct a kind of sequence of attribute and two kinds of sequences of attribute subset to calculate $RE_B(x)$ and $W_B(x)$.

Definition 8: The sequence S of attribute is defined as

$$S = \langle c'_1, c'_2, \dots, c'_m \rangle \quad (11)$$

where $NE(\{c'_k\}) \leq NE(\{c'_{k+1}\})$.

Further starting from the set of single condition attributes (c'_1 or c'_m), each time the attribute is added from S using the forward or reverse way until a set containing all condition attribute is obtained. Thus, the two kinds of sequences of attribute subset can be determined.

Definition 9: The forward sequence FS and reverse sequence RS of attribute subset are defined as, respectively

$$FS = \langle C_1, C_2, \dots, C_m \rangle \quad (12)$$

$$RS = \langle C'_1, C'_2, \dots, C'_m \rangle \quad (13)$$

where $C_j \subseteq C$, $C_m = C$, $C_1 = \{c'_1\}$ and $C_{k+1} = C_k \cup \{c'_{k+1}\}$; $C'_k \subseteq C$, $C'_1 = \{c'_m\}$, $C'_m = C$ and $C'_{k+1} = C'_k \cup \{c'_{m-k}\}$.

From the above Definitions 8 and 9, each attribute set in the three kinds of attribute sequences can determine a neighborhood relation, so that the neighborhood entropy with respect to a neighborhood relation is obtained. Therefore, the three kinds of multigranulation entropy sequences are constructed on the basis of three attribute sequences S , FS, and RS.

Definition 10: The three kinds of multigranulation entropy sequences are defined as, respectively

$$MES = \langle NE(\{c'_1\}), \dots, NE(\{c'_m\}) \rangle \quad (14)$$

$$FMES = \langle NE(C_1), \dots, NE(C_m) \rangle \quad (15)$$

$$RMES = \langle NE(C'_1), \dots, NE(C'_m) \rangle. \quad (16)$$

Based on the above definition, the three kinds of multigranulation relative entropy-based matrices are also constructed.

Definition 11: The three kinds of multigranulation relative entropy-based matrices are defined as, respectively

$$EM_S = \left(RE_{\{c'_k\}}(x_i) \right)_{n \times m} \quad (17)$$

$$EM_{FS} = \left(RE_{C_k}(x_i) \right)_{n \times m} \quad (18)$$

$$EM_{RS} = \left(RE_{C'_k}(x_i) \right)_{n \times m}. \quad (19)$$

Similarly, the three kinds of multigranulation weight function-based matrices can be obtained.

Definition 12: The three kinds of multigranulation weight function-based matrices are defined as, respectively

$$WM_S = \left(W_{\{c'_k\}}(x_i) \right)_{n \times m} \quad (20)$$

$$WM_{FS} = \left(W_{C_k}(x_i) \right)_{n \times m} \quad (21)$$

$$WM_{RS} = \left(W_{C'_k}(x_i) \right)_{n \times m}. \quad (22)$$

Definition 13: The average multigranulation relative entropy and weight function-based matrix are calculated by, respectively

$$AEM = \frac{1}{3}(EM_S + EM_{FS} + EM_{RS}) \quad (23)$$

$$AWM = \frac{1}{3}(WM_S + WM_{FS} + WM_{RS}). \quad (24)$$

The construction of the average multigranulation relative neighborhood-based matrix combines the information of three types of attribute sequences, which lays the foundation for constructing the multigranulation relative entropy-based outlier factor.

In real life, it is more meaningful to assign a certain outlier degree to each object for many complex data scenarios. Therefore, Breunig *et al.* [9] proposed a density-based local outlier identification method. In this article, based on the average multigranulation relative entropy and weight function-based matrix, the MREOF is constructed to indicate the outlier degree of every object.

Definition 14: For $\forall x_i \in U$, the multigranulation relative entropy-based outlier factor $MREOF(x_i)$ of x_i is defined as

$$MREOF(x_i) = 1 - \frac{\sum_{k=1}^m AEM(i, k)AWM(i, k)}{|C|} \quad (25)$$

where $AEM(i, k)$ and $AWM(i, k)$ denote the elements in the i th row and k th column of average multigranulation relative entropy and weight function-based matrix, respectively.

Definition 15: Let μ be a judgement threshold. For $\forall x_i \in U$, if $MREOF(x_i) > \mu$, then x_i is called a multigranulation relative entropy-based outlier.

Example 2 (Continued of Example 1): Let $\delta = 1$, the related descending MREOF values are listed in Table II. Given threshold $\mu = 0.54$, there have $MREOF(x_8) > \mu$ and $MREOF(x_7) > \mu$. Therefore, the outlier set is $OS = \{x_7, x_8\}$. Compared with algorithm NIEOD, algorithm MREOD considers the idea of sequence and multigranulation, so it can accurately detect all objective outliers.

In general, a detection method only gives outlier factors of objects. Before employing a detection method, the user should

TABLE II
DESCENDING MREOF VALUE

U	$MREOF$
x_8	0.5527
x_7	0.5422
x_1	0.4583
x_5	0.3532
x_2	0.3298
x_6	0.2611
x_4	0.2017
x_3	0.1272

Algorithm 1: MREOD

Input: $IS = (U, C, V, f)$, μ and δ .

Output: Outlier set OS .

```

1  $OS \leftarrow \emptyset$ 
2 for  $k \leftarrow 1$  to  $m$  do
3   Calculate  $DN_U(nr_{\{c_k\}})$ 
4   Calculate  $NE(\{c_k\})$ 
5 end
6 Determine  $S = \langle c'_1, c'_2, \dots, c'_m \rangle$ 
7 Construct  $FS = \langle C_1, C_2, \dots, C_m \rangle$  and
    $RS = \langle C'_1, C'_2, \dots, C'_m \rangle$ 
8 for  $k \leftarrow 1$  to  $m$  do
9   Calculate  $DN_U(nr_{C_k})$  and  $DN_U(nr_{C'_k})$ 
10  Calculate  $NE(C_k)$  and  $NE(C'_k)$ 
11 end
12 for  $i \leftarrow 1$  to  $n$  do
13   for  $k \leftarrow 1$  to  $m$  do
14     Calculate  $RE_{\{c_k\}}(x_i)$ ,  $RE_{C_k}(x_i)$ , and  $RE_{C'_k}(x_i)$ 
15     Calculate  $W_{\{c_k\}}(x_i)$ ,  $W_{C_k}(x_i)$ , and  $W_{C'_k}(x_i)$ 
16   end
17   Calculate  $MREOF(x_i)$ 
18   if  $MREOF(x_i) > \mu$  then
19      $OS \leftarrow OS \cup \{x_i\}$ 
20   end
21 end
22 return  $OS$ 

```

first enter an experience value ev to represent expected number of outliers. In our MREOD method, μ can be determined by number ev as follows [35].

- 1) $MREOF(x_i)$ is arranged in descending order to obtain a sequence of objects $x_{i_1}, \dots, x_{i_n}$ in U .
- 2) μ is set up by locating $\mu \in [MREOF(x_{i_{ev+1}}), MREOF(x_{i_{ev}})]$. The method guarantees that ev objects $x_{i_1}, \dots, x_{i_{ev}}$ necessarily hold higher outlier factor in U , and thus, these ev ones are ultimately fed back to the user as outliers.

To further illustrate the above idea, an outlier detection framework is drawn in Fig. 2.

D. Detection Algorithm

In this section, we propose and design a corresponding algorithm (MREOD).

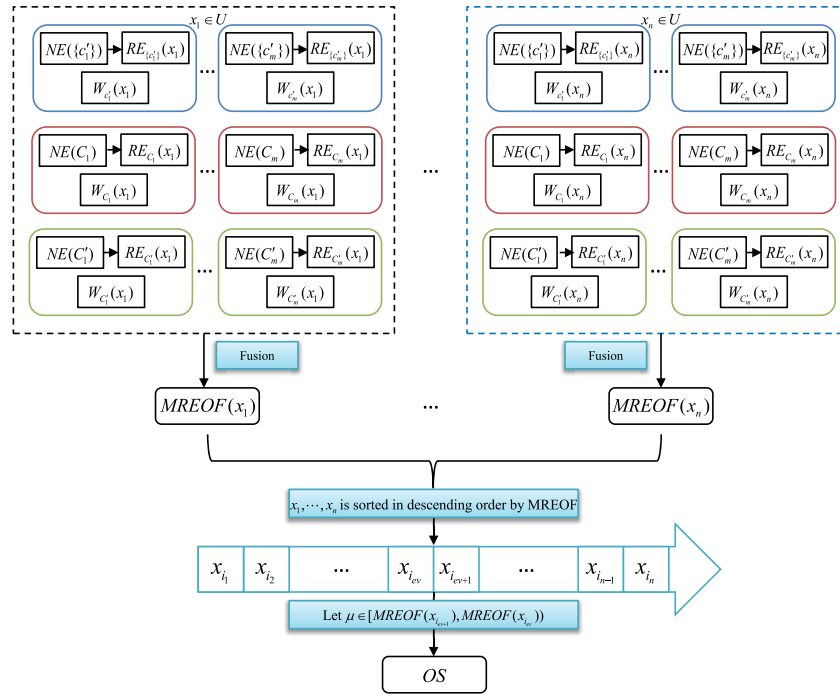


Fig. 2. Framework of outlier detection.

TABLE III
PREPROCESSING AND DESCRIPTION OF DATA SETS

No	Data set	Corresponding IS	Preprocessing	Conditional attribute		Object	Outlier
				Numerical	Categorical		
1	Credit approval	IS_{Cr}	Downsampling the class “+” to 25 objects	6	9	382	25
2	German	IS_{Ge}	Downsampling the class “2” to 14 objects	7	13	714	14
3	Heart disease	IS_{Hd}	Downsampling the class “2” to 16 objects	6	7	166	16
4	Hepatitis	IS_{He}	Downsampling the class “2” to 9 objects	6	13	94	9
5	Diabetes	IS_{Di}	Downsampling the class “tested_positive” to 26 objects	8	0	526	26
6	Ionosphere	IS_{Io}	Downsampling the class “b” to 24 objects	34	0	249	24
7	Iris	IS_{Ir}	Downsampling the class “Iris-virginica” to 11 objects	4	0	111	11
8	Pima	IS_{Pi}	Downsampling the class “TURE” to 55 objects	9	0	555	55
9	Sonar	IS_{So}	Downsampling the class “M” to 10 objects	60	0	107	10
10	Wisconsin diagnostic breast cancer	IS_{Wd}	Downsampling the class “M” to 39 objects	31	0	396	39
11	Wisconsin breast cancer	IS_{Wb}	202 “malignant” (outlier) and 14 “benign” objects were removed [7]	9	0	483	39
12	Yeast	IS_{Ye}	Classes “ERL” (outlier), “CYT”, “NUC”, and “MIT” are selected [25]	8	0	1141	5
13	Lymphography	IS_{Ly}	Classes “1” and “4” are considered as outliers [7]	0	8	148	6
14	Mushroom	IS_{Mu}	Downsampling the class “+” to 221 objects	0	22	4429	221

For algorithm MREOD, the frequency of steps 2–5 is $m \times n \times n$, the frequency of steps 8–11 is m , and the frequency of steps 12–21 is $m \times n$. So the frequency of algorithm MREOD is $m \times n \times n + m + m \times n$. Therefore, in the worst case, the time complexity of algorithm MREOD is $O(mn^2)$.

IV. DATA EXPERIMENTS

In this section, to evaluate the effectiveness of algorithm MREOD, 14 data sets (including categorical, numeric, and mixed attributes) are selected for experiments.¹ In order to form a very unbalanced distribution, the downsampling method proposed in [49] and [50] is used to obtain some data sets for evaluating outlier detection methods. A particular class is randomly downsampled to produce outliers while all objects of the remaining classes are preserved to form an outlier detection data set. In addition, for the missing values of data sets, the maximum probability value method is used

to fill the missing value, that is, the value of attribute with the highest frequency on other objects is used to fill the missing attribute value. In particular, for data set Credit, 37 objects missing one or more values are ignored. The preprocessing and description of data sets are outlined in Table III.²

On these 14 data sets, we compare the performance of algorithm MREOD with the distance-based method (DIS) [51], k-nearest neighbor (kNN)-based method [52], density-based method (LOF) [9], cluster-based method (find cluster-based local outlier factor, FindCBLOF) [7], GrC-based method [19], sequence and RS-based method (SEQ) [20], information entropy (IE)-based method [21], GrC and RST-based method (outlier detection based on GrC and rough set, ODGrCR) [25], and NIEOD [35]. Among them, algorithms DIS, kNN, and LOF are relatively simple and only applicable to numerical attribute data. Algorithm FindCBLOF is well related to clustering, whereas it is only applicable to categorical attribute data.

¹<http://archive.ics.uci.edu/ml>

²<https://github.com/BELLoney/Outlier-detection>

TABLE IV
OPTIMAL PARAMETER SETTING AND DISCRETIZATION METHOD FOR DIFFERENT DATA SETS

IS	$k(kNN)$	$MinPts(LOF)$	$s(FindCBLOF)$	$w(GrC)$	$\lambda(NIEOD)$	$\delta(MREOD)$	Discretization method
IS_{Cr}	33	95	1	3	0.7	0.4	EW
IS_{Ge}	31	135	4	4	1.4	0.4	EW
IS_{Hea}	31	30	5	4	0.9	1.0	EW
IS_{Hep}	16	8	8	4	0.3	0.4	EW
IS_{Di}	7	76	7	3	1.2	1.6	EW
IS_{Io}	2	2	4	5	0.3	1.0	EW
IS_{Ir}	13	2	1	3	1.2	0.5	EW
IS_{Pi}	136	109	6	5	0.9	1.1	EW
IS_{So}	13	19	2	2	0.8	0.9	EW
IS_{Wd}	74	41	2	2	0.6	0.7	EW
IS_{Wb}	39	15	3 [7]	2 [19]	0.4	0.5	—
IS_{Ye}	243	8	5	3	0.1	0.1	EW
IS_{Ly}	37	36	3 [7]	2 [19]	—	—	—
IS_{Mu}	212	2	9	10	—	—	—

Algorithms GrC, SEQ, IE, and ODGrCR are outlier detection algorithms with rough set as the framework. They are only applicable to categorical attribute data, while for numerical attribute data, discretization preprocessing is required. Algorithm NIEOD is suitable for mixed attribute data, which only considers the information about single attribute.

A. Experimental Setup

In our experiments, for algorithms MREOD and NIEOD, parameter λ and δ need to be set, respectively. We calculate the optimal values for parameter λ and δ in the range $[0.1, 2]$ with the step size 0.1. We repeat algorithms kNN and LOF for 14 data sets and calculated the optimal values of their respective parameters k and $MinPts$ in the range $[1, n/4]$ with the step size 1. Two parameters α and β required by algorithm FindCBLOF are set to 90% and 5, respectively [7]. The optimal values s for algorithm FindCBLOF is calculated in the range $[1, 10]$ with the step size 1. For algorithm GrC, the overlap distance metric is used to calculate the distance between any two objects [19], whose parameter $d = |C|/w$. We calculate w 's integer optimal values in the range $[1, 10]$. In addition, for algorithms SEQ, IE, and ODGrCR, all conditional attribute values in IS_{Wb} and IS_{Ly} are considered to be categorical type [7]. For the remaining data sets with numerical attributes, discretization methods based on equal width (EW) and equal frequency (EF) are used to convert all numerical attribute values into discrete forms with three interval numbers, and the discretization method with most excellent result is finally taken. For algorithms DIS, kNN, and LOF, the Euclidean distance metric is adopted. All different categorical attribute values are replaced with different integer values, and then all attribute values are normalized to $[0, 1]$ interval by using the min-max normalization. Moreover, an outlier is treated as a binary property in the traditional DIS-based method. Obviously, it is unreasonable. Thus, the strategy of [21] is used to define a distance outlier factor to represent the outlier degree for every object. Finally, the optimal parameter settings and discretization methods for different data sets are described in Table IV.

For all algorithms, their performances are estimated by the evaluation method introduced by Aggarwal and Yu [10] and

Yuan *et al.* [35]. It mainly includes “top ratio (number of objects)” and “number of outlier included (coverage ratio).” First, ten outlier detection algorithms are used to calculate the outlier factor for each object. Then, all objects are sorted in ascending order through $MREOF(x_i)$. Then, let OS_{top} denote the sets of top t objects, and OS_{true} denote the sets of true outliers in U . So, for each algorithm, “number of objects” is equal to $|OS_{top}| = t$, and “number of true outlier included” is equal to $|OS_{inc}| = |OS_{top} \cap OS_{true}|$, while the top ratio TR and coverage ratio CR are calculated as follows, respectively

$$TR = \frac{|OS_{top}|}{|U|} = \frac{t}{|U|} \quad (26)$$

$$CR = \frac{|OS_{top} \cap OS_{true}|}{|OS_{true}|} = \frac{|OS_{inc}|}{|OS_{true}|} \quad (27)$$

Obviously, the higher CR, the better the algorithm works. Using this method, MREOD's parameter just needs to set δ , which is used to adjust the neighborhood radius on the basis of the standard deviation.

B. Mixed Attribute Data Set

As shown in Table V, the CR of algorithm MREOD is higher than that of the other algorithms on most mixed attribute data sets.

For example, for IS_{Cr} , when the t is 43 (the TR is 11.26%), $|OS_{inc}|$ detected by algorithm MREOD is 25. But for algorithms DIS, kNN, LOF, FindCBLOF, GrC, SEQ, IE, ODGrCR, and NIEOD, only 20, 19, 15, 19, 21, 22, 22, 20, and 21 outliers are found, respectively. In terms of IS_{Hea} and IS_{Hep} , the same results can be analyzed.

Algorithm MREOD considers the integration of attribute sequences. While algorithm NIEOD only considers the information of a single attribute. Therefore, algorithm MREOD should theoretically be superior to algorithm NIEOD. However, algorithm MREOD is slightly weaker than algorithm NIEOD in IS_{Ge} , which may be caused by abnormal data distribution.

In comparative experiments, algorithm MREOD involves only one parameter δ . The fluctuation of δ will lead to the change of CR calculated by algorithm. The variation of CR

TABLE V
COMPARATIVE EXPERIMENTAL RESULTS ON MIXED ATTRIBUTE DATA SETS

IS	$t(TR(\%))$	$ OS^{inc} (CR(\%))$									
		DIS	kNN	LOF	FindCBLOF	GrC	SEQ	IE	ODGrCR	NIEOD	MREOD
IS_{Cr}	13(3.06)	12(48.00)	9(36.00)	9(36.00)	12(48.00)	11(44.00)	10(40.00)	12(48.00)	11(44.00)	13(52.00)	13(52.00)
	25(5.88)	17(68.00)	15(60.00)	10(40.00)	15(60.00)	16(64.00)	18(72.00)	17(68.00)	17(68.00)	17(68.00)	18(72.00)
	43(10.12)	20(80.00)	19(76.00)	15(60.00)	19(76.00)	21(84.00)	22(88.00)	22(88.00)	20(80.00)	21(84.00)	25(100.00)
	64(15.06)	20(80.00)	21(84.00)	17(68.00)	22(88.00)	23(92.00)	25(100.00)	24(96.00)	22(88.00)	23(92.00)	25(100.00)
	67(15.76)	21(84.00)	21(84.00)	19(76.00)	23(92.00)	23(92.00)	25(100.00)	24(96.00)	23(92.00)	25(100.00)	25(100.00)
	78(18.35)	22(88.00)	21(84.00)	19(76.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	23(92.00)	25(100.00)	25(100.00)
	82(19.29)	22(88.00)	21(84.00)	19(76.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)
	110(25.88)	23(92.00)	25(100.00)	21(84.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)
	145(34.12)	25(100.00)	25(100.00)	21(84.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)
IS_{Ge}	291(68.47)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)	25(100.00)
	7(0.98)	2(14.29)	4(28.57)	2(14.29)	3(21.43)	3(21.43)	2(14.29)	3(21.43)	3(21.43)	3(21.43)	3(21.43)
	14(1.96)	3(21.43)	4(28.57)	3(21.43)	5(35.71)	5(35.71)	5(35.71)	4(28.57)	5(35.71)	4(28.57)	5(35.71)
	35(4.90)	5(35.71)	8(57.14)	5(35.71)	7(50.00)	7(50.00)	9(64.29)	8(57.14)	10(71.43)	14(100.00)	10(71.43)
	44(6.16)	6(42.86)	9(64.29)	6(42.86)	10(71.43)	10(71.43)	9(64.29)	11(78.57)	13(92.86)	14(100.00)	14(100.00)
	54(7.56)	9(64.29)	9(64.29)	9(64.29)	14(100.00)	14(100.00)	10(71.43)	13(92.86)	14(100.00)	14(100.00)	14(100.00)
	65(9.10)	9(64.29)	11(78.57)	9(64.29)	14(100.00)	14(100.00)	10(71.43)	14(100.00)	14(100.00)	14(100.00)	14(100.00)
	80(11.20)	11(78.57)	14(100.00)	11(78.57)	14(100.00)	14(100.00)	10(71.43)	14(100.00)	14(100.00)	14(100.00)	14(100.00)
	121(16.95)	14(100.00)	14(100.00)	12(85.71)	14(100.00)	14(100.00)	12(85.71)	14(100.00)	14(100.00)	14(100.00)	14(100.00)
IS_{Hea}	127(17.79)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	12(85.71)	14(100.00)	14(100.00)	14(100.00)	14(100.00)
	174(24.37)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)	14(100.00)
	8(4.82)	8(50.00)	8(50.00)	7(43.75)	8(50.00)	8(50.00)	7(43.75)	7(43.75)	7(43.75)	8(50.00)	7(43.75)
	16(9.64)	12(75.00)	12(75.00)	11(68.75)	14(87.50)	15(93.75)	11(68.75)	13(81.25)	13(81.25)	14(87.50)	15(93.75)
	17(10.24)	13(81.25)	13(81.25)	11(68.75)	15(93.75)	15(93.75)	11(68.75)	13(81.25)	14(87.50)	15(93.75)	16(100.00)
	18(10.84)	13(81.25)	13(81.25)	12(75.00)	15(93.75)	15(93.75)	12(75.00)	14(87.50)	15(93.75)	15(93.75)	16(100.00)
	19(11.45)	13(81.25)	13(81.25)	13(81.25)	15(93.75)	15(93.75)	12(75.00)	15(93.75)	15(93.75)	16(100.00)	16(100.00)
	30(18.07)	14(87.50)	14(87.50)	14(87.50)	15(93.75)	15(93.75)	15(93.75)	16(100.00)	16(100.00)	16(100.00)	16(100.00)
	32(19.28)	14(87.50)	16(100.00)	14(87.50)	16(100.00)	16(100.00)	15(93.75)	15(93.75)	16(100.00)	16(100.00)	16(100.00)
IS_{Hep}	34(20.48)	14(87.50)	16(100.00)	14(87.50)	16(100.00)	16(100.00)	15(93.75)	16(100.00)	16(100.00)	16(100.00)	16(100.00)
	36(21.69)	14(87.50)	16(100.00)	14(87.50)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)
	39(23.49)	14(87.50)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)
	43(25.90)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)	16(100.00)
	5(5.32)	4(44.44)	4(44.44)	4(44.44)	5(55.56)	5(55.56)	4(44.44)	5(55.56)	5(55.56)	4(44.44)	4(44.44)
	9(9.57)	8(88.89)	6(66.67)	7(77.78)	8(88.89)	7(77.78)	5(55.56)	7(77.78)	7(77.78)	8(88.89)	8(88.89)
	10(10.64)	8(88.89)	7(77.78)	8(88.89)	8(88.89)	7(77.78)	6(66.67)	8(88.89)	7(77.78)	9(100.00)	9(100.00)
	11(11.70)	8(88.89)	8(88.89)	9(100.00)	8(88.89)	8(88.89)	7(77.78)	8(88.89)	8(88.89)	9(100.00)	9(100.00)
	12(12.77)	8(88.89)	8(88.89)	9(100.00)	8(88.89)	8(88.89)	8(88.89)	8(88.89)	9(100.00)	9(100.00)	9(100.00)
IS_{Hep}	13(13.83)	9(100.00)	9(100.00)	9(100.00)	8(88.89)	8(88.89)	8(88.89)	9(100.00)	9(100.00)	9(100.00)	9(100.00)
	14(14.89)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	8(88.89)	9(100.00)	9(100.00)	9(100.00)	9(100.00)
	15(15.96)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)	9(100.00)

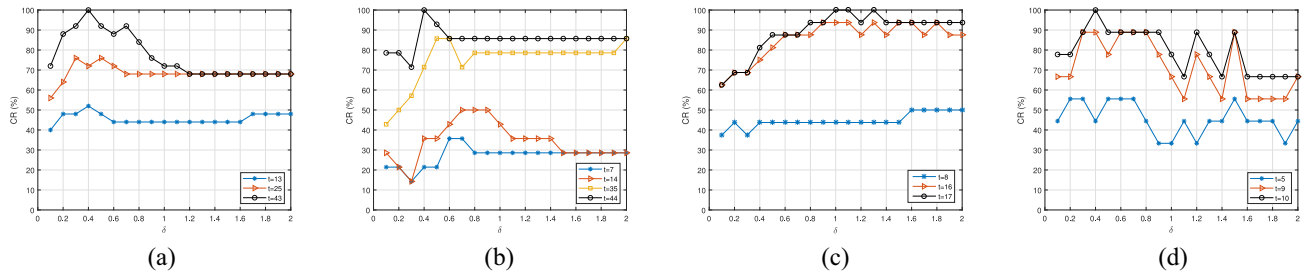


Fig. 3. Line charts of CR with change of δ on mixed data sets. (a) IS_{Cr} . (b) IS_{Ge} . (c) IS_{Hea} . (d) IS_{Hep} .

with δ is shown in Fig. 3, which contains some top kinds level of TR in Table V.

For example, based on Fig. 3(a), we can analyze the change of CR when the level of TR is 11.26% (t is equal to 43). Specifically, for algorithm MREOD, its experimental results generally have increased then decreased with the increase of δ . Moreover, algorithm MREOD achieves the best result if $\delta = 0.4$, that is, the number of outliers detected is the most one. As shown in Fig. 3(a), when choosing the other levels, similar results can be obtained for algorithm MREOD. The number of outliers detected by algorithm MREOD is larger than that of other nine algorithms in the most range of δ .

In conclusion, compared with other nine algorithms, algorithm MREOD has better performance for mixed attribute data

sets. Therefore, we can see that it can be effectively used in mixed attribute data.

C. Numerical Attribute Data Set

The experimental results on numerical attribute data set in Table VI show that the performance of algorithm MREOD is significantly better than that of algorithms NIEOD, ODGrCR, IE, SEQ, GrC, FindCBLOF, LOF, DIS, and kNN.

For algorithm MREOD, the variation of CR with the change of δ change is depicted in Fig. 4. It also contains some top kinds level of TR in Table VI.

For example, for the level of TR is 9.11% (t is equal to 44) in Fig. 4(g), when $\delta \geq 0.5$, algorithm MREOD

TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS ON NUMERICAL ATTRIBUTE DATA SETS

IS	$t(TR(\%))$	$ OS^{inc} (CR(\%))$									
		DIS	kNN	LOF	FindCBLOF	GrC	SEQ	IE	ODGrCR	NIEOD	MREOD
IS_{Di}	16(3.04)	9(34.62)	7(26.92)	10(38.46)	6(23.08)	9(34.62)	10(38.46)	10(38.46)	6(23.08)	12(46.15)	12(46.15)
	26(4.94)	11(42.31)	12(46.15)	12(46.15)	9(34.62)	12(46.15)	10(38.46)	12(46.15)	10(38.46)	18(69.23)	16(61.54)
	44(8.37)	15(57.69)	17(65.38)	17(65.38)	15(57.69)	18(69.23)	12(46.15)	17(65.38)	17(65.38)	25(96.15)	26(100.00)
	46(8.75)	16(61.54)	17(65.38)	17(65.38)	15(57.69)	19(73.08)	13(50.00)	17(65.38)	18(69.23)	26(100.00)	26(100.00)
	81(15.40)	24(92.31)	26(100.00)	24(92.31)	22(84.62)	24(92.31)	23(88.46)	23(88.46)	24(92.31)	26(100.00)	26(100.00)
	83(15.78)	24(92.31)	26(100.00)	26(100.00)	22(84.62)	25(96.15)	23(88.46)	24(92.31)	24(92.31)	26(100.00)	26(100.00)
	91(17.30)	26(100.00)	26(100.00)	26(100.00)	24(92.31)	25(96.15)	24(92.31)	24(92.31)	24(92.31)	26(100.00)	26(100.00)
	106(20.15)	26(100.00)	26(100.00)	26(100.00)	24(92.31)	26(100.00)	24(92.31)	25(96.15)	25(96.15)	26(100.00)	26(100.00)
	108(20.53)	26(100.00)	26(100.00)	26(100.00)	24(92.31)	26(100.00)	24(92.31)	26(100.00)	26(100.00)	26(100.00)	26(100.00)
	117(22.24)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	24(92.31)	26(100.00)	26(100.00)	26(100.00)	26(100.00)
	226(42.97)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)	26(100.00)
IS_{Io}	12(4.82)	12(50.00)	12(50.00)	12(50.00)	11(45.83)	11(45.83)	11(45.83)	10(41.67)	12(50.00)	12(50.00)	12(50.00)
	24(9.64)	22(91.67)	24(100.00)	22(91.67)	18(75.00)	18(75.00)	18(75.00)	17(70.83)	20(83.33)	23(95.83)	24(100.00)
	26(10.44)	23(95.83)	24(100.00)	22(91.67)	19(79.17)	19(79.17)	18(75.00)	18(75.00)	21(87.50)	24(100.00)	24(100.00)
	31(12.45)	24(100.00)	24(100.00)	23(95.83)	20(83.33)	20(83.33)	20(83.33)	19(79.17)	24(100.00)	24(100.00)	24(100.00)
	48(19.28)	24(100.00)	24(100.00)	24(100.00)	23(95.83)	23(95.83)	23(95.83)	23(95.83)	24(100.00)	24(100.00)	24(100.00)
	52(20.88)	24(100.00)	24(100.00)	24(100.00)	23(95.83)	23(95.83)	23(95.83)	24(100.00)	24(100.00)	24(100.00)	24(100.00)
	125(50.20)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)	24(100.00)
IS_{Ir}	6(5.41)	6(54.55)	5(45.45)	6(54.55)	6(54.55)	6(54.55)	6(54.55)	2(18.18)	6(54.55)	6(54.55)	6(54.55)
	11(9.91)	11(100.00)	8(72.73)	11(100.00)	10(90.91)	9(81.82)	9(81.82)	7(63.64)	9(81.82)	11(100.00)	11(100.00)
	12(10.81)	11(100.00)	9(81.82)	11(100.00)	11(100.00)	10(90.91)	10(90.91)	8(72.73)	10(90.91)	11(100.00)	11(100.00)
	13(11.71)	11(100.00)	10(90.91)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	9(81.82)	11(100.00)	11(100.00)	11(100.00)
	14(12.61)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	10(90.91)	11(100.00)	11(100.00)	11(100.00)
	15(13.51)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)	11(100.00)
IS_{Pi}	17(3.06)	11(20.00)	12(21.82)	11(20.00)	10(18.18)	12(21.82)	12(21.82)	12(21.82)	12(21.82)	12(21.82)	13(23.64)
	25(4.50)	13(23.64)	14(25.45)	14(25.45)	14(25.45)	16(29.09)	14(25.45)	15(27.27)	15(27.27)	19(34.55)	18(32.73)
	55(9.91)	27(49.09)	29(52.73)	27(49.09)	31(56.36)	27(49.09)	22(40.00)	26(47.27)	28(50.91)	35(63.64)	38(69.09)
	87(15.68)	41(74.55)	46(83.64)	40(72.73)	37(67.27)	42(76.36)	27(49.09)	35(63.64)	45(81.82)	55(100.00)	51(92.73)
	96(17.30)	45(81.82)	47(85.45)	44(80.00)	39(70.91)	44(80.00)	28(50.91)	37(67.27)	45(81.82)	55(100.00)	55(100.00)
	145(26.13)	55(100.00)	54(98.18)	55(100.00)	53(96.36)	52(94.55)	33(60.00)	49(89.09)	52(94.55)	55(100.00)	55(100.00)
	165(29.73)	55(100.00)	55(100.00)	55(100.00)	53(96.36)	52(94.55)	34(61.82)	54(98.18)	52(94.55)	55(100.00)	55(100.00)
	175(31.53)	55(100.00)	55(100.00)	55(100.00)	53(96.36)	52(94.55)	34(61.82)	55(100.00)	52(94.55)	55(100.00)	55(100.00)
	180(32.43)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	54(98.18)	34(61.82)	55(100.00)	52(94.55)	55(100.00)	55(100.00)
	181(32.61)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	34(61.82)	55(100.00)	52(94.55)	55(100.00)	55(100.00)
	187(33.69)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	34(61.82)	55(100.00)	55(100.00)	55(100.00)	55(100.00)
	369(66.49)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)	55(100.00)
IS_{So}	5(4.67)	4(40.00)	4(40.00)	5(50.00)	4(40.00)	5(50.00)	5(50.00)	5(50.00)	4(40.00)	5(50.00)	5(50.00)
	10(9.35)	8(80.00)	7(70.00)	8(80.00)	8(80.00)	8(80.00)	8(80.00)	8(80.00)	8(80.00)	8(80.00)	8(80.00)
	12(11.21)	8(80.00)	8(80.00)	9(90.00)	10(100.00)	8(80.00)	9(90.00)	8(80.00)	9(90.00)	10(100.00)	10(100.00)
	13(12.15)	9(90.00)	9(90.00)	10(100.00)	10(100.00)	9(90.00)	9(90.00)	8(80.00)	9(90.00)	10(100.00)	10(100.00)
	14(13.08)	9(90.00)	10(100.00)	10(100.00)	10(100.00)	9(90.00)	9(90.00)	8(80.00)	9(90.00)	10(100.00)	10(100.00)
	15(14.02)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	9(90.00)	9(90.00)	10(100.00)	10(100.00)	10(100.00)
	18(16.82)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	9(90.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)
	22(20.56)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)	10(100.00)
IS_{Wd}	19(4.80)	17(43.59)	17(43.59)	18(46.15)	18(46.15)	19(48.72)	10(25.64)	18(46.15)	18(46.15)	18(46.15)	18(46.15)
	39(9.85)	35(89.74)	33(84.62)	36(92.31)	37(94.87)	37(94.87)	21(53.85)	37(94.87)	37(94.87)	37(94.87)	37(94.87)
	41(10.35)	36(92.31)	35(89.74)	38(97.44)	38(97.44)	38(97.44)	23(58.97)	37(94.87)	38(97.44)	39(100.00)	39(100.00)
	42(10.61)	37(94.87)	36(92.31)	39(100.00)	38(97.44)	38(97.44)	24(61.54)	38(97.44)	38(97.44)	39(100.00)	39(100.00)
	43(10.86)	38(97.44)	37(94.87)	39(100.00)	38(97.44)	39(100.00)	25(64.10)	38(97.44)	38(97.44)	39(100.00)	39(100.00)
	45(11.36)	38(97.44)	39(100.00)	39(100.00)	38(97.44)	39(100.00)	27(69.23)	38(97.44)	38(97.44)	39(100.00)	39(100.00)
	46(11.62)	39(100.00)	39(100.00)	39(100.00)	38(97.44)	39(100.00)	28(71.79)	38(97.44)	39(100.00)	39(100.00)	39(100.00)
	49(12.37)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	31(79.49)	39(100.00)	39(100.00)	39(100.00)	39(100.00)
	64(16.16)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)
IS_{Wb}	20(4.14)	20(51.28)	20(51.28)	20(51.28)	17(43.59)	17(43.59)	18(46.15)	18(46.15)	18(46.15)	20(51.28)	20(51.28)
	39(8.07)	34(87.18)	34(87.18)	35(89.74)	31(79.49)	31(79.49)	32(82.05)	32(82.05)	33(84.62)	34(87.18)	35(89.74)
	44(9.11)	38(97.44)	36(92.31)	36(92.31)	34(87.18)	34(87.18)	34(87.18)	34(87.18)	37(94.87)	36(92.31)	39(100.00)
	49(10.14)	38(97.44)	38(97.44)	39(100.00)	35(89.74)	37(94.87)	36(92.31)	36(92.31)	37(94.87)	39(100.00)	39(100.00)
	50(10.35)	38(97.44)	39(100.00)	39(100.00)	35(89.74)	37(94.87)	37(94.87)	37(94.87)	37(94.87)	39(100.00)	39(100.00)
	52(10.77)	39(100.00)	39(100.00)	39(100.00)	36(92.31)	38(97.44)	38(97.44)	37(94.87)	38(97.44)	39(100.00)	39(100.00)
	54(11.18)	39(100.00)	39(100.00)	39(100.00)	37(94.87)	38(97.44)	38(97.44)	38(97.44)	39(100.00)	39(100.00)	39(100.00)
	56(11.59)	39(100.00)	39(100.00)	39(100.00)	38(97.44)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)
	64(13.25)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)	39(100.00)
IS_{Ye}	3(0.26)	0(0.00)	3(60.00)	3(60.00)	0(0.00)	2(40.00)	0(0.00)	1(20.00)	0(0.00)	2(40.00)	3(60.00)
	5(0.44)	1(20.00)	5(100.00)	5(100.00)	2(40.00)	2(40.00)	2(40.00)	2(40.00)	2(40.00)	3(60.00)	5(100.00)
	6(0.53)	1(20.00)	5(100.00)	5(100.00)	3(60.00)	3(60.00)	2(40.00)	2(40.00)	3(60.00)	4(80.00)	5(100.00)
	7(0.61)	1(20.00)	5(100.00)	5(100.00)	4(80.00)	4(80.00)	2(40.00)	2(40.00)	4(80.00)	5(100.00)	5(100.00)
	8(0.70)	1(20.00)	5(100.00)	5(100.00)	5(100.00)	4(80.00)	2(40.00)	2(40.00)	4(80.00)	5(100.00)	5(100.00)
	9(0.79)	1(20.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	2(40.00)	2(40.00)	4(80.00)	5(100.00)	5(100.00)
	10(0.88)	2(40.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	3(60.00)	2(40.00)	5(100.00)	5(100.00)	5(100.00)
	13(1.14)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	3(60.00)	5(100.00)	5(100.00)	5(100.00)
	23(2.02)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)	5(100.00)

generally has stability and optimization in IS_{Wb} . Algorithm MREOD gets the largest number of outliers included when $\delta = 0.5$.

In summary, the comparative experimental results show that algorithm MREOD also has a good effect on numerical attribute data sets.

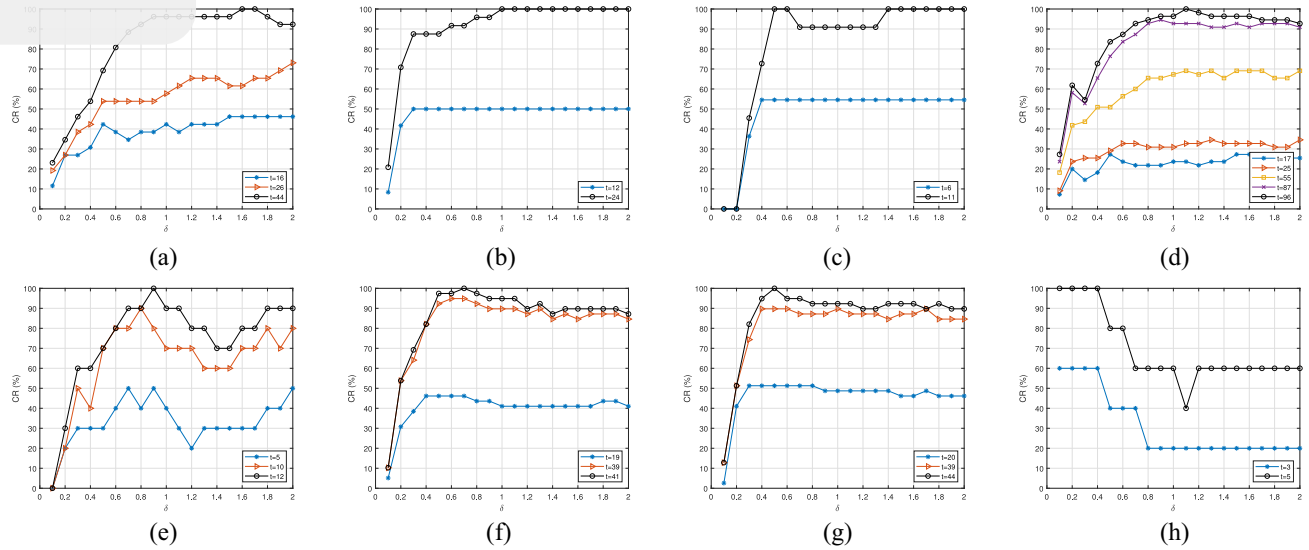


Fig. 4. Line charts of CR with change of δ on numerical attribute data sets. (a) IS_{Di} . (b) IS_{Io} . (c) IS_{Ir} . (d) IS_{Pi} . (e) IS_{So} . (f) IS_{Wd} . (g) IS_{Wb} . (h) IS_{Ye} .

TABLE VII
COMPARATIVE EXPERIMENTAL RESULTS ON CATEGORICAL ATTRIBUTE DATA SETS

IS	$t(TR(\%))$	$ OS^{inc} (CR(\%))$									
		DIS	kNN	LOF	FindCBLOF	GrC	SEQ	IE	ODGrCR	NIEOD	MREOD
IS_{Ly}	3(2.03)	3(50.00)	3(50.00)	2(33.33)	2(33.33)	3(50.00)	3(50.00)	3(50.00)	3(50.00)	3(50.00)	3(50.00)
	6(4.05)	5(83.33)	4(66.67)	4(66.67)	3(50.00)	5(83.33)	4(66.67)	5(83.33)	5(83.33)	5(83.33)	6(100.00)
	9(6.08)	5(83.33)	6(100.00)	5(83.33)	4(66.67)	6(100.00)	5(83.33)	6(100.00)	6(100.00)	6(100.00)	6(100.00)
	10(6.76)	6(100.00)	6(100.00)	6(100.00)	4(66.67)	6(100.00)	5(83.33)	6(100.00)	6(100.00)	6(100.00)	6(100.00)
	12(8.11)	6(100.00)	6(100.00)	6(100.00)	4(66.67)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)
	30(20.27)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)	6(100.00)
IS_{Mu}	110(2.48)	34(15.38)	47(21.27)	44(19.91)	99(44.80)	108(48.87)	108(48.87)	110(49.77)	110(49.77)	110(49.77)	72(32.58)
	221(4.99)	44(19.91)	49(22.17)	46(20.81)	193(87.33)	193(87.33)	193(87.33)	189(85.52)	193(87.33)	193(87.33)	171(77.38)
	756(17.07)	212(95.93)	106(47.96)	197(89.14)	193(87.33)	195(88.24)	195(88.24)	201(90.95)	214(96.83)	199(90.05)	221(100.00)
	1011(22.83)	217(98.19)	157(71.04)	206(93.21)	217(98.19)	196(88.69)	199(90.05)	217(98.19)	221(100.00)	205(92.76)	221(100.00)
	1185(26.76)	220(99.55)	221(100.00)	214(96.83)	220(99.55)	199(90.05)	199(90.05)	217(98.19)	221(100.00)	215(97.29)	221(100.00)
	1231(27.79)	220(99.55)	221(100.00)	214(96.83)	221(100.00)	199(90.05)	199(90.05)	217(98.19)	221(100.00)	219(99.10)	221(100.00)
	1340(30.26)	220(99.55)	221(100.00)	214(96.83)	221(100.00)	199(90.05)	199(90.05)	219(99.10)	221(100.00)	221(100.00)	221(100.00)
	1342(30.30)	221(100.00)	221(100.00)	214(96.83)	221(100.00)	199(90.05)	199(90.05)	219(99.10)	221(100.00)	221(100.00)	221(100.00)
	1440(32.51)	221(100.00)	221(100.00)	214(96.83)	221(100.00)	199(90.05)	199(90.05)	221(100.00)	221(100.00)	221(100.00)	221(100.00)
	2143(48.39)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	215(97.29)	219(99.10)	221(100.00)	221(100.00)	221(100.00)	221(100.00)
	2171(49.02)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	217(98.19)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)
	2337(52.77)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)	221(100.00)

D. Categorical Attribute Data Set

The comparative experimental results on categorical attribute data set are shown in Table VII. From Table VII, we can conclude that the performance of algorithm MREOD is better than that of the other nine algorithms. The comparative experimental results show that algorithm MREOD is also applicable to categorical data set.

V. CONCLUSION

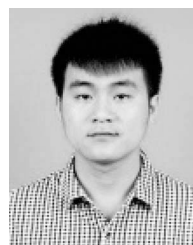
In this article, we propose a mixed attribute outlier detection method based on multigranulation relative entropy. Based on the new NIS and neighborhood entropy, the idea of sequence and multigranulation is fused to build a outlier detection model. As a result, this method makes up for the shortcomings of the existing methods, which is suitable for categorical, numerical, and mixed attribute data set. The comparative experimental results show that the proposed method has better adaptability and effectiveness. Few people study outlier detection method using neighborhood rough set techniques. The

work of this article enriches the application of neighborhood rough set in data mining and provides a novel method for mixed attribute outlier detection. In future work, in order to effectively process dynamic data, incremental outlier detection methods can be further studied.

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